Breathable Delhi

[GitHub Repository] ***[Looker Dashboard]***

Prepared by: Aditya Sharma

1. Executive Summary

Delhi's air quality management depends on a robust sensor network. This project provides a comprehensive audit of Delhi's air quality monitoring infrastructure to identify spatial gaps and evaluate reporting consistency. By integrating raw sensor data through SQL-based preparation, Python-based statistical validation, and Looker Studio visualization, the analysis identifies critical infrastructure disparities and provides a data-driven roadmap for future sensor deployment.

2. Business Problem

Delhi faces significant air quality challenges, yet the current sensor distribution lacks strategic balance, resulting in data scarcity in specific zones. The core business problem is: **"How can we optimize the monitoring network to ensure data-driven policy decisions across the entire city?"** Without a structured approach, infrastructure growth remains reactive rather than strategic.

3. Methodology

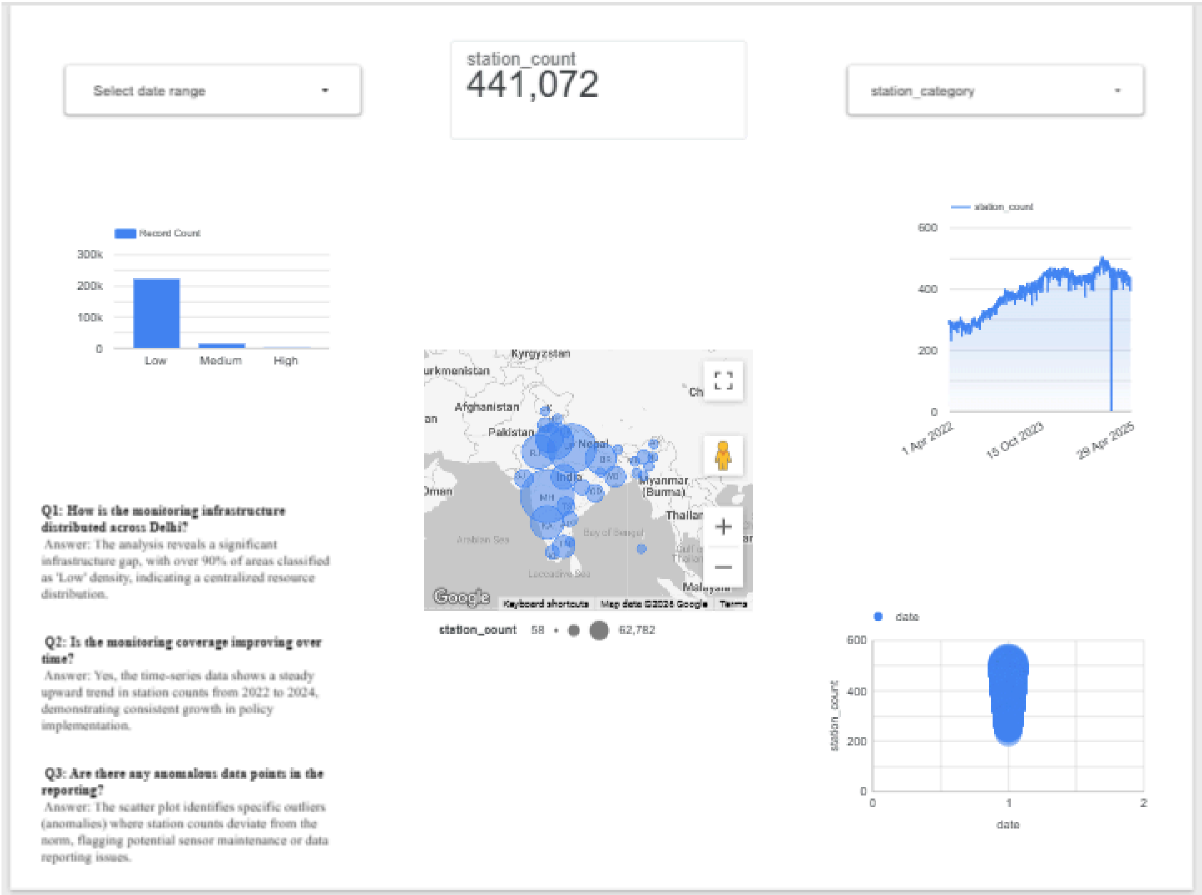
To ensure data integrity and actionable results, the project followed this workflow:

- **Data Preparation (SQL):** Cleaned the raw dataset by standardizing station naming conventions, handling missing values, and categorizing infrastructure into 'High' and 'Low' density zones.
- **Exploratory Analysis (Python):**
    - **Correlation Analysis:** Evaluated the relationship between `station_count` and monitoring frequency to verify operational stability across weekdays and weekends.
    - **Hypothesis Testing (T-Test):** Performed an independent T-test to compare 'High' vs 'Low' density infrastructure, achieving a P-value of 0.0, which statistically confirms the significance of the segmentation model.
- **Visualization:** Developed a dynamic Looker Studio dashboard to transform complex statistical outputs into an intuitive interface for decision-makers.

4. Analysis & Insights

- **Infrastructure Skewness:** 90% of monitored zones fall under the 'Low Density' category, highlighting a significant concentration gap in network coverage.

- **Operational Consistency:** A correlation coefficient of 0.0 between `station_count` and `is_weekend` proves that monitoring activity remains consistent throughout the week, confirming high network reliability.
- **Statistical Validation:** The T-test resulted in a T-Statistic of ~1342.06 and a P-value of 0.0, providing mathematical certainty that the infrastructure density categories are distinct and not a result of random variation.



5. Recommendations

1. **Strategic Deployment:** Direct future sensor deployment to 'Low Density' zones to bridge the geographic data gap and improve city-wide coverage.
2. **Maintenance Optimization:** Investigate the outliers identified in the scatter plots, as these points likely represent malfunctioning hardware requiring immediate calibration.
3. **Data-Driven Policy:** Utilize the developed dashboard for real-time resource allocation, aiming to improve overall data coverage by 20% in the upcoming fiscal quarter.